

Create a study guide for the following nmap flags. Include definition. Include output of each nmap flag. Include a detailed explanation of each element for each nmap output. Do not reference citations in the explanation! -sS → TCP SYN scan (stealth scan) -sT → TCP connect scan -sU → UDP scan -sA → ACK scan (firewall detection) -sN → Null scan -sF → FIN scan -sX → Xmas scan

Below is a compact study guide for each requested Nmap flag. For each scan type you'll see:

1. Definition / purpose
 2. Typical command and sample output
 3. Detailed explanation of every element in that output (line by line, column by column)
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1. -sS → TCP SYN Scan (Stealth / Half-open)

1.1 Definition

- Sends a TCP SYN packet to each target port.
- If SYN/ACK comes back → port is considered **open**.
- If RST comes back → port is **closed**.
- If nothing / ICMP error → often **filtered**.
- Nmap then **does not complete** the TCP handshake (doesn't send ACK), so it's "half-open" and somewhat stealthier than a full connect scan.

1.2 Example Command

```
nmap -sS 192.168.1.10
```

1.3 Sample Output

```
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-04 10:00 UTC
Nmap scan report for 192.168.1.10
Host is up (0.0013s latency).
Not shown: 997 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
```

Nmap done: 1 IP address (1 host up) scanned in 0.23 seconds

1.4 Explanation of Output

Line 1: Starting Nmap 7.94 ... at 2026-04-04 10:00 UTC

- Starting Nmap 7.94 – Version of Nmap used for the scan.
- <https://nmap.org> – Project URL; purely informational.
- at 2026-04-04 10:00 UTC – Start time of the scan with timezone (UTC). Useful for logging and correlation.

Line 2: Nmap scan report for 192.168.1.10

- Nmap scan report for – Indicates a new host section.
- 192.168.1.10 – Target host; may be shown as hostname or IP (if DNS resolution is enabled and a name exists).

Line 3: Host is up (0.0013s latency).

- Host is up – Nmap successfully probed the host (e.g., via ping / ARP / other host discovery method) and it responded.
- (0.0013s latency) – Round-trip time between Nmap and the host (approx 1.3 ms here). Gives you a feel for how “far” the host is in network terms.

Line 4: Not shown: 997 closed ports

- Not shown: – Nmap hides ports not deemed interesting according to default settings.
- 997 closed ports – Out of the default 1,000 scanned, 997 responded with RST, indicating they are closed.

Table Header: PORT STATE SERVICE

- PORT – Port number and protocol.
- STATE – Nmap’s classification of the port’s state.
- SERVICE – Nmap’s guess of the common service typically bound to that port.

Row: 22/tcp open ssh

- 22/tcp – Port 22 using TCP.

- **open** – Target sent SYN/ACK back in response to our SYN; something is listening.
- **ssh** – Nmap maps port 22 to the SSH service from its internal service database.

Row: 80/tcp open http

- **80/tcp** – Port 80 TCP.
- **open** – SYN/ACK received.
- **http** – Common usage for port 80 (web server).

Row: 443/tcp open https

- **443/tcp** – Port 443 TCP.
- **open** – SYN/ACK received.
- **https** – Common usage for encrypted web traffic.

Last line: Nmap done: 1 IP address (1 host up) scanned in 0.23 seconds

- **1 IP address** – Total target IPs considered.
 - **(1 host up)** – Number of responsive hosts.
 - **scanned in 0.23 seconds** – Total elapsed scan time.
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2. -sT → TCP Connect Scan

2.1 Definition

- Uses the OS's **full TCP connect()** call instead of raw packets.
- Performs the complete 3-way handshake (SYN → SYN/ACK → ACK).
- Requires no raw-socket privileges; works for unprivileged users.
- Easier to detect (logs look like normal connections).

2.2 Example Command

`nmap -sT 192.168.1.10`

2.3 Sample Output

```
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-04 10:05 UTC
Nmap scan report for 192.168.1.10
Host is up (0.0011s latency).
Not shown: 996 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
3306/tcp   open  mysql
```

Nmap done: 1 IP address (1 host up) scanned in 0.34 seconds

2.4 Explanation

This output is structurally identical to `-sS`; what differs in reality is **how** Nmap decided each port state:

- 22/tcp open ssh, 80/tcp open http, 443/tcp open https, 3306/tcp open mysql:
 - Here, for each port, Nmap actually completed the TCP handshake using `connect()`.
 - Application logs on the target are more likely to show these as real incoming connections.

Interpretation of each line/column is the same as in `-sS`; only the scanning technique differs behind the scenes.

3. `-sU` → UDP Scan

3.1 Definition

- Sends UDP datagrams to target ports instead of TCP.
- If it receives an **ICMP “port unreachable”** → port is **closed**.
- If it gets **no response** → port is **open|filtered** (could be open or silently dropped by a firewall).
- If it gets a valid UDP response → port is **open**.
- UDP scanning is often **slow** because there is no handshake and many ports just don’t respond.

3.2 Example Command

```
nmap -sU 192.168.1.10
```

3.3 Sample Output

```
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-04 10:10 UTC
Nmap scan report for 192.168.1.10
Host is up (0.0015s latency).
Not shown: 997 closed udp ports
PORT      STATE      SERVICE
53/udp    open       domain
123/udp   open       ntp
161/udp   open|filtered snmp
```

Nmap done: 1 IP address (1 host up) scanned in 7.81 seconds

3.4 Explanation

Lines 1–3: same meaning as before (Nmap version, host, latency).

Line 4: Not shown: 997 closed udp ports

- Emphasizes that this is a **UDP** scan (not TCP).
- **closed udp ports** – Nmap got ICMP “port unreachable” messages for 997 ports.

Header: PORT STATE SERVICE

- **PORT** – Port and protocol (/udp).
- **STATE** – Note the special **open|filtered** state can appear here (UDP ambiguity).
- **SERVICE** – Guessed common UDP service.

Row: 53/udp open domain

- **53/udp** – Port 53 UDP.
- **open** – A valid UDP response came back (e.g., DNS reply).
- **domain** – Typical service name for DNS.

Row: 123/udp open ntp

- **123/udp** – Port 123 UDP.
- **open** – Target responded (e.g., NTP response).

- `ntp` – Network Time Protocol.

Row: 161/udp `open|filtered snmp`

- 161/udp – Port 161 UDP.
- `open|filtered` – No response at all; could mean:
 - Service is open but not replying to that probe, **or**
 - Packet is dropped by a firewall.
- `snmp` – Commonly used UDP port for SNMP.

Last line: as before, but note the **longer scan time (7.81s)** indicating UDP scans are slower.

4. `-sA` → ACK Scan (Firewall / Ruleset Detection)

4.1 Definition

- Sends **ACK** packets (not SYN) to target ports.
- Used primarily to **map firewall rules**, not to find open services.
- If a port responds with **RST** → port is considered **unfiltered** (firewall is not blocking the ACK).
- If no response / ICMP error → port is **filtered**.
- Ports are usually labeled **unfiltered** or **filtered**, not **open** / **closed**.

4.2 Example Command

```
nmap -sA 192.168.1.10
```

4.3 Sample Output

```
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-04 10:15 UTC
Nmap scan report for 192.168.1.10
Host is up (0.0010s latency).
PORT      STATE      SERVICE
22/tcp    unfiltered ssh
80/tcp    unfiltered http
443/tcp   filtered  https
```

```
Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
```

4.4 Explanation

Lines 1–3: same structure as before.

Header: PORT STATE SERVICE

- PORT – Ports probed with ACKs.
- STATE – Focus on firewall behavior:
 - **unfiltered** – ACK packet passed through (RST came back).
 - **filtered** – No usable response (likely blocked).

Row: 22/tcp **unfiltered** **ssh**

- 22/tcp – Port 22 TCP.
- **unfiltered** – ACK got a RST back → firewall allowed the probe; host reachable at that port.
- **ssh** – Typical SSH service.

Row: 80/tcp **unfiltered** **http**

- Same meaning for HTTP on port 80.

Row: 443/tcp **filtered** **https**

- 443/tcp – Port 443.
- **filtered** – ACK probes did not receive RST; likely blocked by firewall rules.
- **https** – Common HTTPS service.

Key point: This scan **does not tell you if the port is open or closed**, only whether a firewall is filtering that port.

5. -sN → Null Scan

5.1 Definition

- Sends TCP packets with **no flags set** (all bits 0).
- Exploits RFC-defined behavior on some TCP stacks:
 - **Closed** ports → must reply with **RST**.
 - **Open** ports → should **ignore** the packet (no response).

- If no response → often labeled **open|filtered** (could be open, or a firewall dropped it).
- Often used as a **stealthy** scan to evade older firewalls/IDS.

5.2 Example Command

```
nmap -sN 192.168.1.10
```

5.3 Sample Output

```
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-04 10:20 UTC
Nmap scan report for 192.168.1.10
Host is up (0.0012s latency).
Not shown: 997 closed ports
PORT      STATE      SERVICE
22/tcp    open|filtered ssh
80/tcp    open|filtered http
443/tcp   closed      https
```

Nmap done: 1 IP address (1 host up) scanned in 0.27 seconds

5.4 Explanation

Line 4: Not shown: 997 closed ports

- 997 ports responded with RST to the null probe → considered closed.

Row: 22/tcp open|filtered ssh

- **open|filtered** – No response to the null packet.
 - Could be an open port that silently ignored the invalid packet, **or**
 - A firewall that dropped the packet.

Row: 80/tcp open|filtered http

- Same reasoning for port 80.

Row: 443/tcp closed https

- **closed** – Target responded with RST to the null packet, indicating the port is closed according to the RFC behavior.

6. -sF → FIN Scan

6.1 Definition

- Sends TCP packets with only the **FIN** flag set (no SYN).
- Similar logic to null scan:
 - Closed ports → send **RST**.
 - Open ports → ignore → **open|filtered**.
- Again, used as a stealthy technique against older filtering devices.

6.2 Example Command

```
nmap -sF 192.168.1.10
```

6.3 Sample Output

```
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-04 10:25 UTC
Nmap scan report for 192.168.1.10
Host is up (0.0014s latency).
Not shown: 996 closed ports
PORT      STATE      SERVICE
22/tcp    open|filtered ssh
80/tcp    open|filtered http
443/tcp   closed     https
8080/tcp   open|filtered http-proxy
```

Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds

6.4 Explanation

Row: 22/tcp open|filtered ssh, 80/tcp open|filtered http, 8080/tcp open|filtered http-proxy

- All of these ports did **not** send a RST in response to the FIN packet.
- Ambiguous result: could be an open port ignoring unusual packets, or a firewall dropping them.

Row: 443/tcp closed https

- Port 443 sent RST → Nmap flags it as **closed**.

The table structure and state meanings match the null scan; the difference is only the flag Nmap sets (FIN instead of no flags).

7. -sX → Xmas Scan

7.1 Definition

- Sends TCP packets with **FIN, PSH, and URG flags set** – “lit up” like a Christmas tree.
- Same RFC-based logic:
 - Closed ports → RST.
 - Open ports → ignore.
- Another classic stealth scan type, historically used to evade old firewalls/IDS.

7.2 Example Command

```
nmap -sX 192.168.1.10
```

7.3 Sample Output

```
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-04 10:30 UTC
Nmap scan report for 192.168.1.10
Host is up (0.0016s latency).
Not shown: 995 closed ports
PORT      STATE      SERVICE
22/tcp    open|filtered ssh
80/tcp    open|filtered http
443/tcp    closed     https
445/tcp    open|filtered microsoft-ds
8080/tcp   open|filtered http-proxy
```

```
Nmap done: 1 IP address (1 host up) scanned in 0.32 seconds
```

7.4 Explanation

Row examples:

- 22/tcp open|filtered ssh, 80/tcp open|filtered http, 445/tcp open|filtered microsoft-ds, 8080/tcp open|filtered http-proxy
 - All ports that did **not** send RST to the Xmas packet.
 - Labeled open|filtered for the same reason as null/FIN scans.
- 443/tcp closed https
 - RST in response → considered closed.

Again, same output structure; what differs is the unusual combination of flags in the probe packets.

How to Study These Scan Types

When preparing for the exam:

- **Memorize the mapping:**
 - `-sS` = SYN/half-open
 - `-sT` = `connect()`
 - `-sU` = UDP
 - `-sA` = ACK (firewall rules)
 - `-sN` = Null
 - `-sF` = FIN
 - `-sX` = Xmas
 - **Remember port states per scan:**
 - SYN/connect: `open`, `closed`, `filtered`.
 - UDP: `open`, `closed`, `open|filtered`.
 - ACK: `unfiltered`, `filtered`.
 - Null/FIN/Xmas: `closed`, `open|filtered`.
 - Practice reading Nmap tables until you can quickly explain each column and state from memory.
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